

Blockhouse QR Project

conceptual questions

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1 What’s the motivation behind measuring OFI at multiple depth levels of the order book?

1. **Enhanced Information Capture (Richer Liquidity Signal).** Analyzing OFI across multiple order book levels enables a richer liquidity signal beyond the top of the book. Limit orders placed deeper in the book often reflect strategic intent or hidden liquidity, which can materially influence price evolution, especially during periods of volatility or large trades
2. **Diurnal-Adjusted Comparability.** To make OFIs at different depths comparable throughout the trading day, the paper scales each level- m OFI by the average depth across the top M levels imbalances are not spuriously large simply because those levels hold more resting volume.
3. **Strong Correlation Among Levels (Common Signal).** Empirically, OFIs across different depths are highly correlated, and the first principal component explains over 89 % of their total variance. This indicates that aggregated multi-level OFI contains meaningful, non-redundant information.
4. **Dimensionality Reduction & Overfitting Control.** By using PCA to integrate multi-level OFIs into a single “Integrated OFI,” the model reduces dimensionality and noise, mitigating the risk of overfitting. This is especially important when constructing sparse, interpretable models in high-frequency trading environments, and it outperforms best-level OFI alone in contemporaneous price-impact models
5. **Enhanced Explanatory & Predictive Power.** The integrated OFI provides better explanatory power for contemporaneous price changes than models based solely on best-level OFI (higher R^2 in price-impact regressions). Moreover, lagged cross-asset OFIs—particularly those aggregated from multiple levels—improve short-term return forecasts (1–5 minutes ahead), although the benefit decays rapidly, highlighting the importance of high-frequency responsiveness.

2 Why do the authors use Lasso regression rather than OLS for estimating cross-impact?

1. **Sparsity of Cross-Impact Effects.** In a high-dimensional, multi-asset setting only a small subset of cross-asset OFIs truly drives any given stock’s return. LASSO enforces sparsity by shrinking many coefficients exactly to zero, automatically selecting the most influential cross-impact terms.
2. **Simultaneous Variable Selection & Regularization.** By penalizing the absolute magnitude of each coefficient, LASSO performs variable selection and regularization in one step. This is essential when the number of candidate predictors (e.g. 100+ cross-asset OFIs) exceeds the number of observations in each minute-level window, thereby avoiding overfitting.
3. **Robustness to Multicollinearity.** Order-flow imbalances across different assets are often highly correlated, which makes OLS estimates unstable. The LASSO penalty shrinks correlated coefficients toward zero, yielding a more stable and interpretable model under multicollinearity.

4. **Improved Predictive Accuracy.** By selecting only impactful variables, LASSO enhances out-of-sample predictive performance—demonstrated by higher R^2 and lower forecasting error compared to OLS in both in-sample and out-of-sample tests.
5. **Empirical Validation of Sparsity.** The authors empirically find that only a handful of assets exert significant contemporaneous cross-impact on each stock, validating the sparsity assumption and making LASSO more appropriate than OLS for uncovering meaningful cross-asset relationships.

3 Why is OFI considered a better predictor of short-term returns than trade volume?

1. **Direct Signal of Supply–Demand Imbalance.** OFI measures the net pressure of buyer- versus seller-initiated orders, whereas trade volume simply aggregates all trades without direction, obscuring whether buying or selling dominates — making OFI a more precise indicator of imminent price moves.
2. **Higher Predictive Power over Short Horizons.** The paper shows that high-frequency OFI (1–5 min bins) has strong explanatory power for subsequent returns, reflecting immediate liquidity imbalances that often precede price changes. Trade volume lacks this directional immediacy and granularity.
3. **Captures Order-Book Depth and Structure.** By integrating OFI across multiple LOB levels, the metric incorporates hidden liquidity and strategic order placement deeper in the book. Trade volume, by contrast, ignores depth and cannot distinguish between superficial and structural liquidity shifts.
4. **Empirical Validation Across Studies.** Both Cont et al. (2014) and the current paper demonstrate that OFI (especially the Integrated OFI) explains a significantly larger fraction of return variance than volume-based models, in-sample and out-of-sample.
5. **Directional Ambiguity of Volume.** High volume can arise from both aggressive buying and selling, making it a noisy gauge of net pressure. OFI resolves this ambiguity by sign-adjusting order flows, yielding a clear measure of market sentiment.